

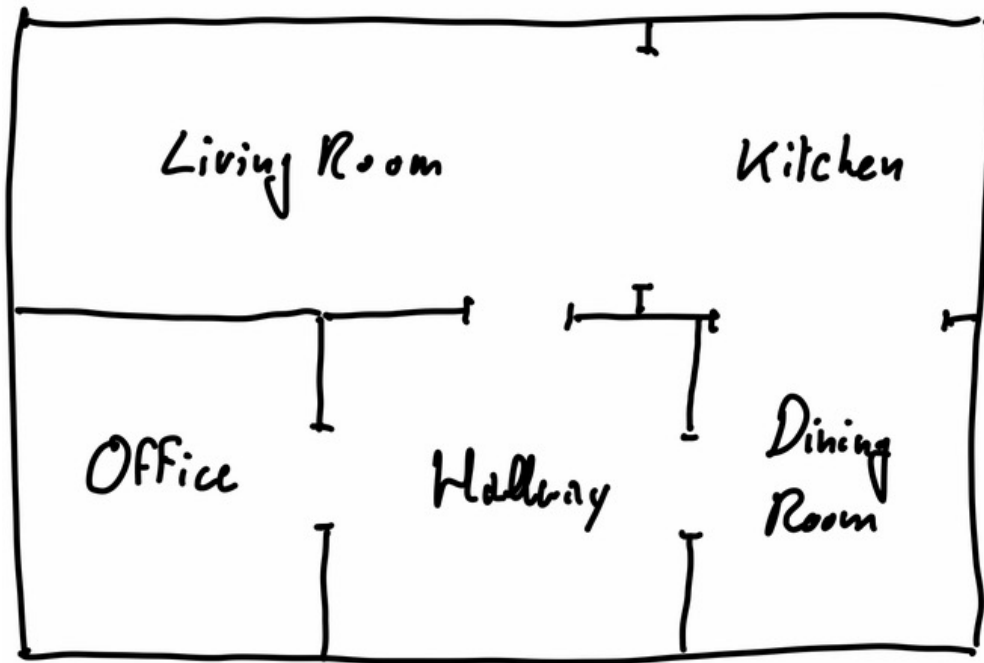
Lecture 9: Sensing & perception

Frank Dellaert • Seth Hutchinson



Five rooms produce just three possible light readings.

SOUTH SIDE: usually brighter



bright

Living room · Kitchen

medium

Office · Dining room

dark

Hallway

Typical light levels, sometimes we get different readings.

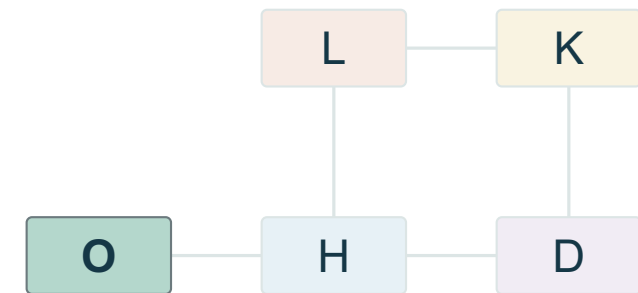
A sensor model predicts a reading given the room—not the other way around.

Current room	dark	medium	bright
Living room	0.1	0.1	0.8
Kitchen	0.1	0.1	0.8
Office	0.2	0.7	0.1
Hallway	0.8	0.1	0.1
Dining room	0.1	0.8	0.1

$$P(Z_t | X_t)$$



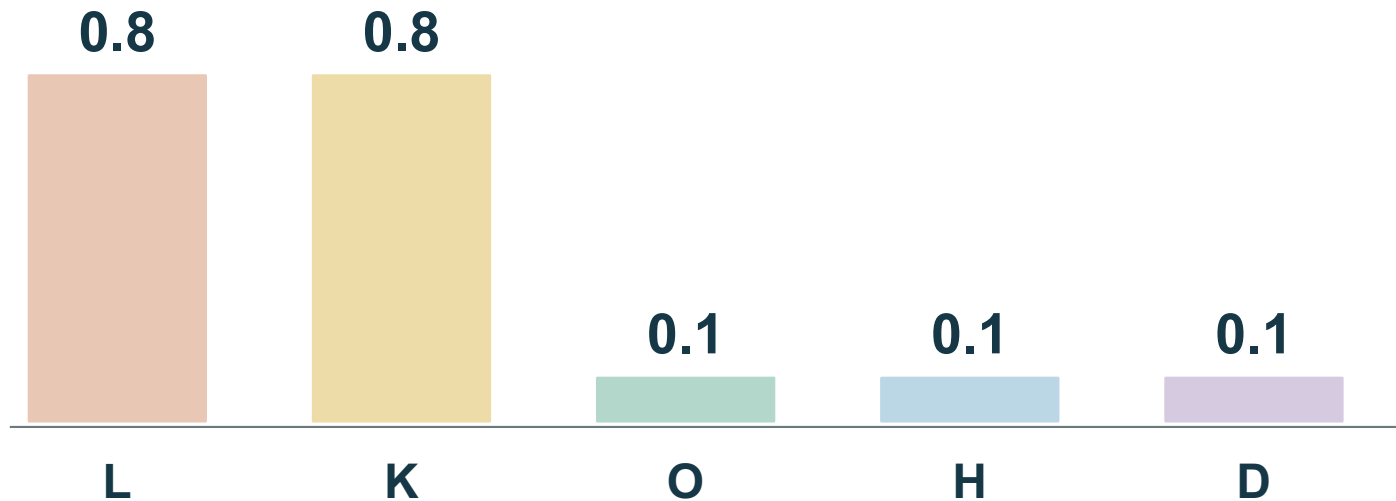
CURRENT ROOM: OFFICE



Choose one room; read its distribution across the three possible measurements.

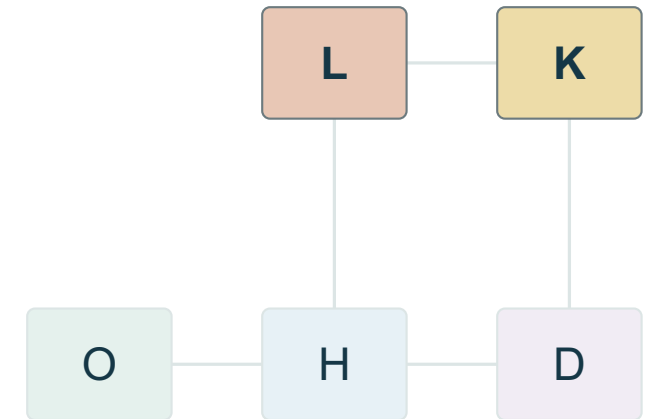
The same reading can come from more than one room.

LIKELIHOOD OF A BRIGHT READING



$$P(\text{bright} \mid L) = P(\text{bright} \mid K) = 0.8$$

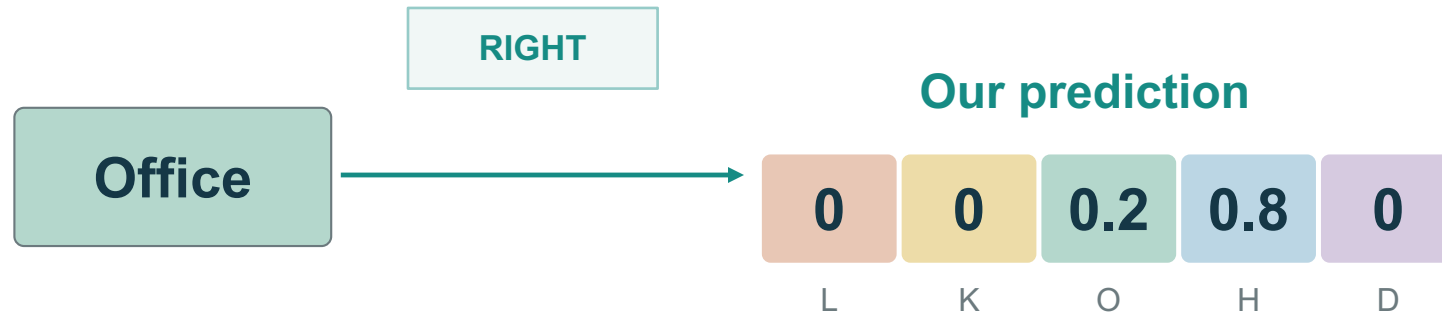
TWO BRIGHT ROOMS



Several rooms are *likely* given a single reading.

We can predict where we might be.
Now we want evidence of where we are.

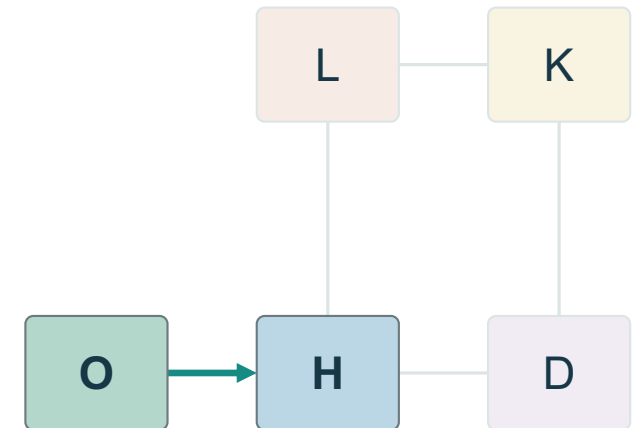
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Office	0.2	0.7	0.1
Hallway	0.8	0.1	0.1
Dining room	0.1	0.8	0.1



$$b_1 = b_0 M_R$$

dark

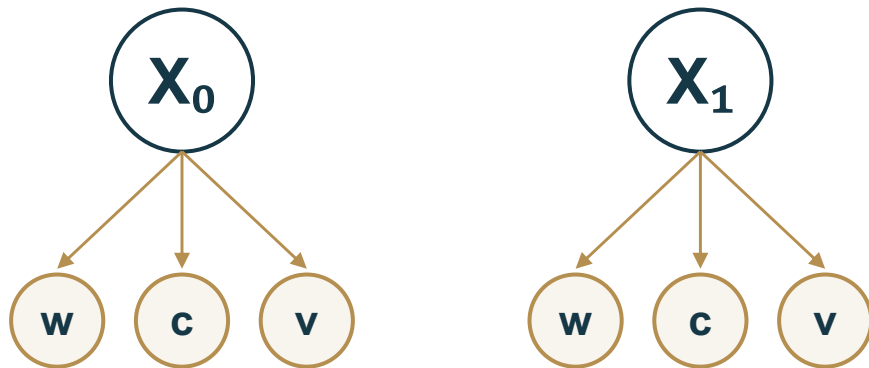
MAP REMINDER



Exercise: What is the posterior probability?

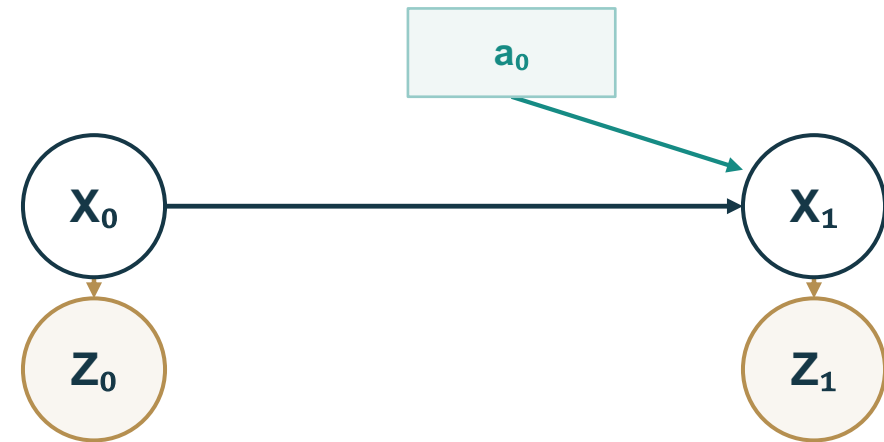
Unlike separate pieces of trash, one robot links measurements across time.

TRASH SORTING: separate items



No temporal link between the two items.

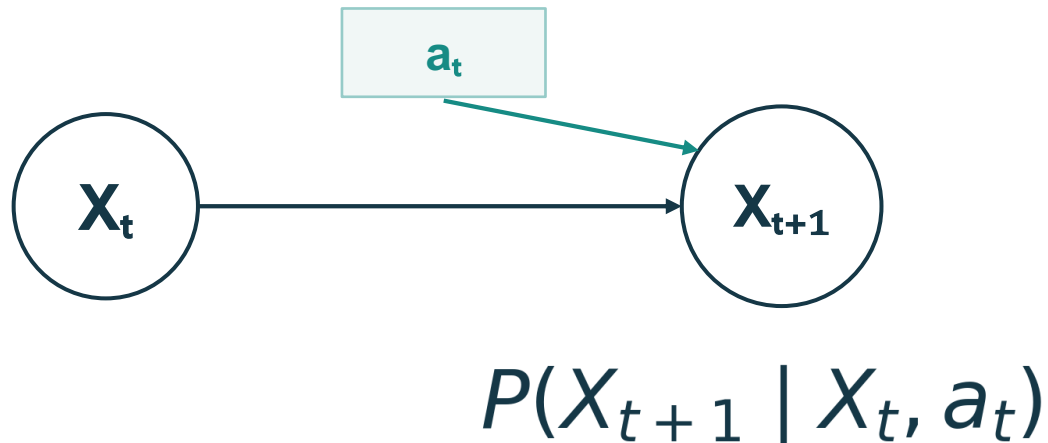
VACUUMING: one moving robot



A motion model links consecutive states.

A node's parents tell us which conditional probability to use.

MOTION MODEL

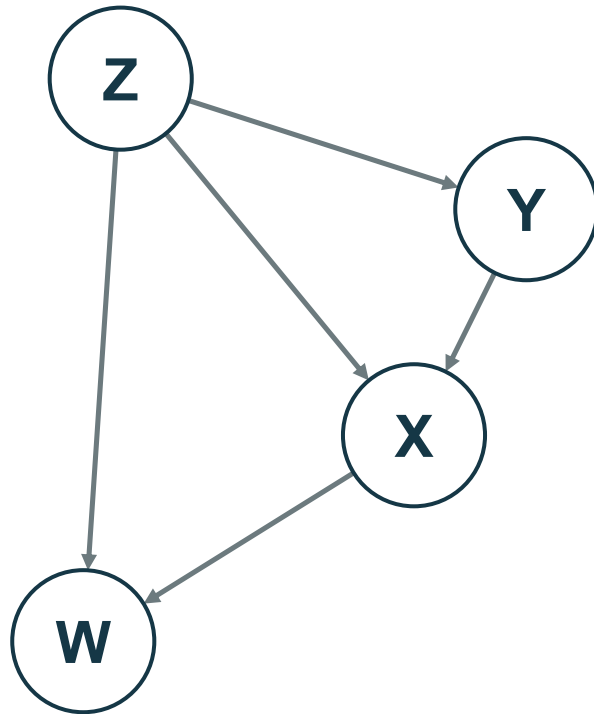


SENSOR MODEL



Know quantities are indicated with a square node

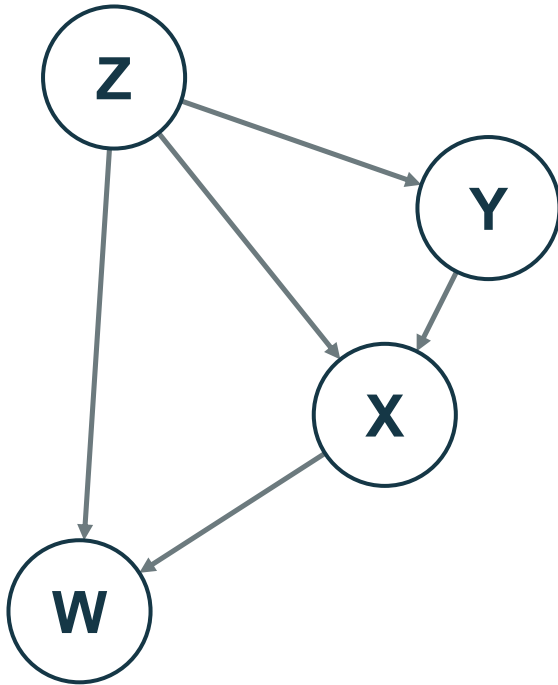
A *Bayes net* is a directed graph with no directed cycles.



	No parents
	Parent: Z
	Parents: Y, Z
	Parents: X, Z

Directed acyclic graph = DAG. Each node stores a local probability distribution.

Multiply one conditional per node to obtain the joint distribution.



$$P(Z)$$

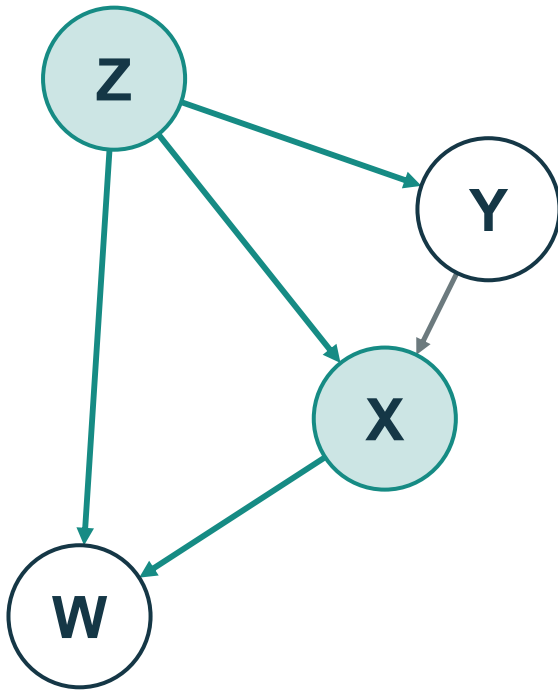
$$P(Y \mid Z)$$

$$P(X \mid Y, Z)$$

$$P(W \mid X, Z)$$

$$P(W, X, Y, Z) = P(W \mid X, Z)P(X \mid Y, Z)P(Y \mid Z)P(Z)$$

Given all of W's parents, Y adds no information about W.



CONDITION ON BOTH PARENTS

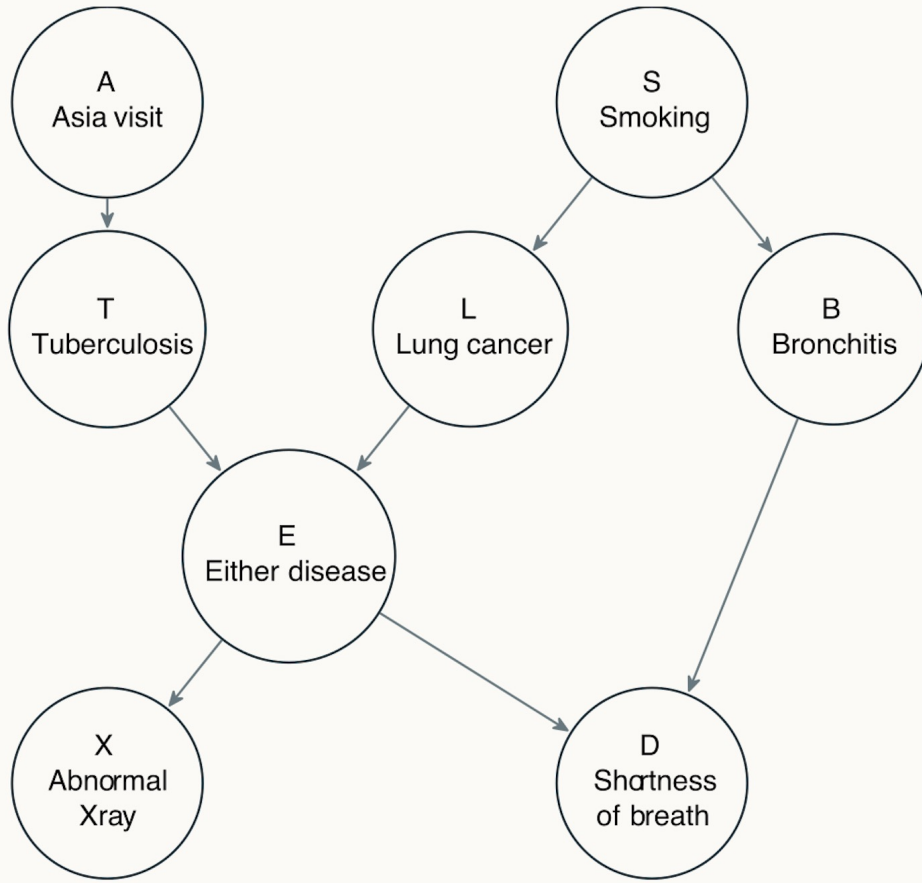


$$P(W \mid X, Y, Z) = P(W \mid X, Z)$$

The conditioning set is $\{X, Z\}$, not X alone.

Asia: a Simplified Medical Diagnosis Example.

Asia network



Conditional probability table

$$P(T \mid A)$$

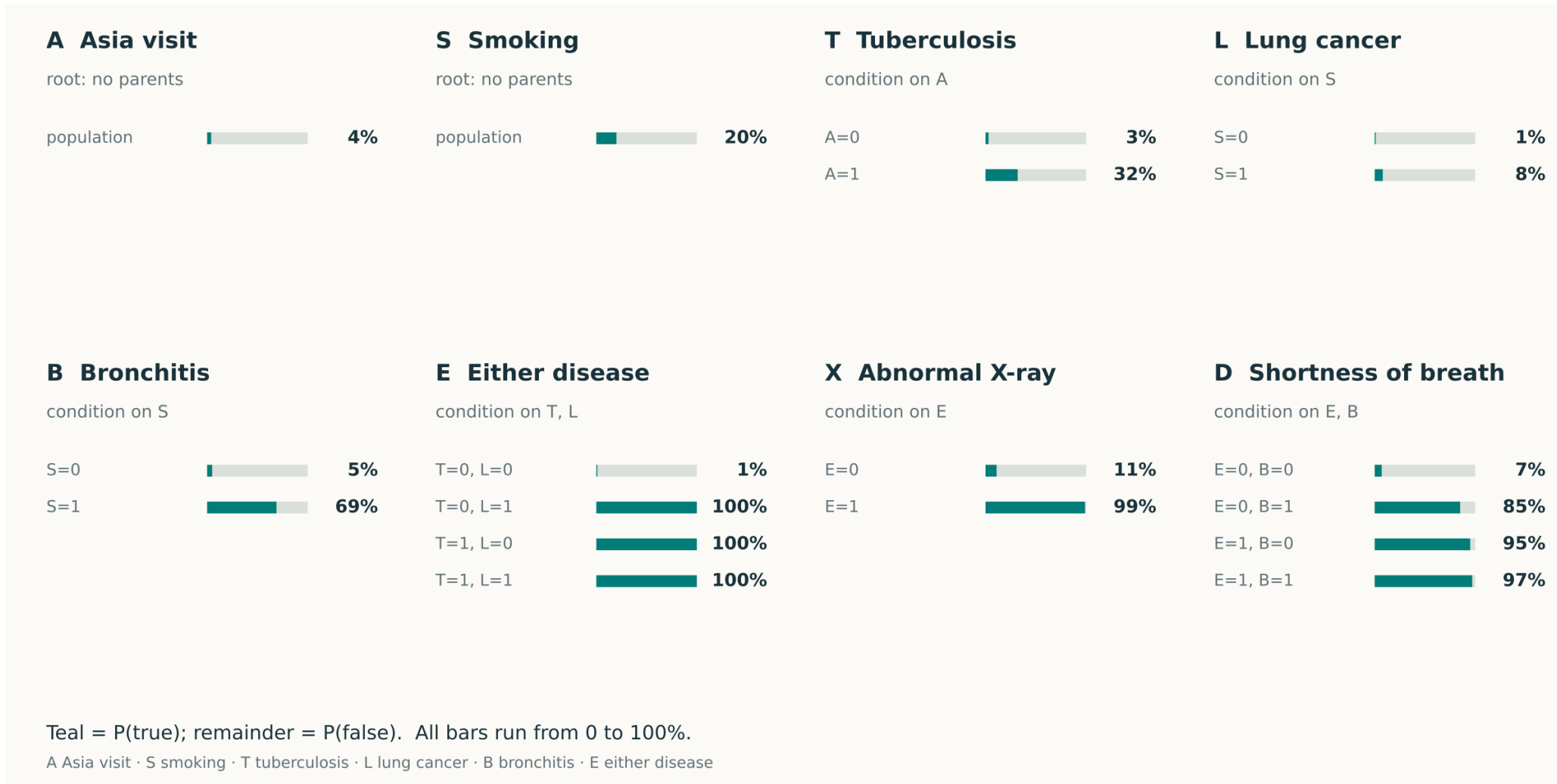
Asia visit	T = false	T = true
No	97%	3%
Yes	68%	32%

Equivalent factor

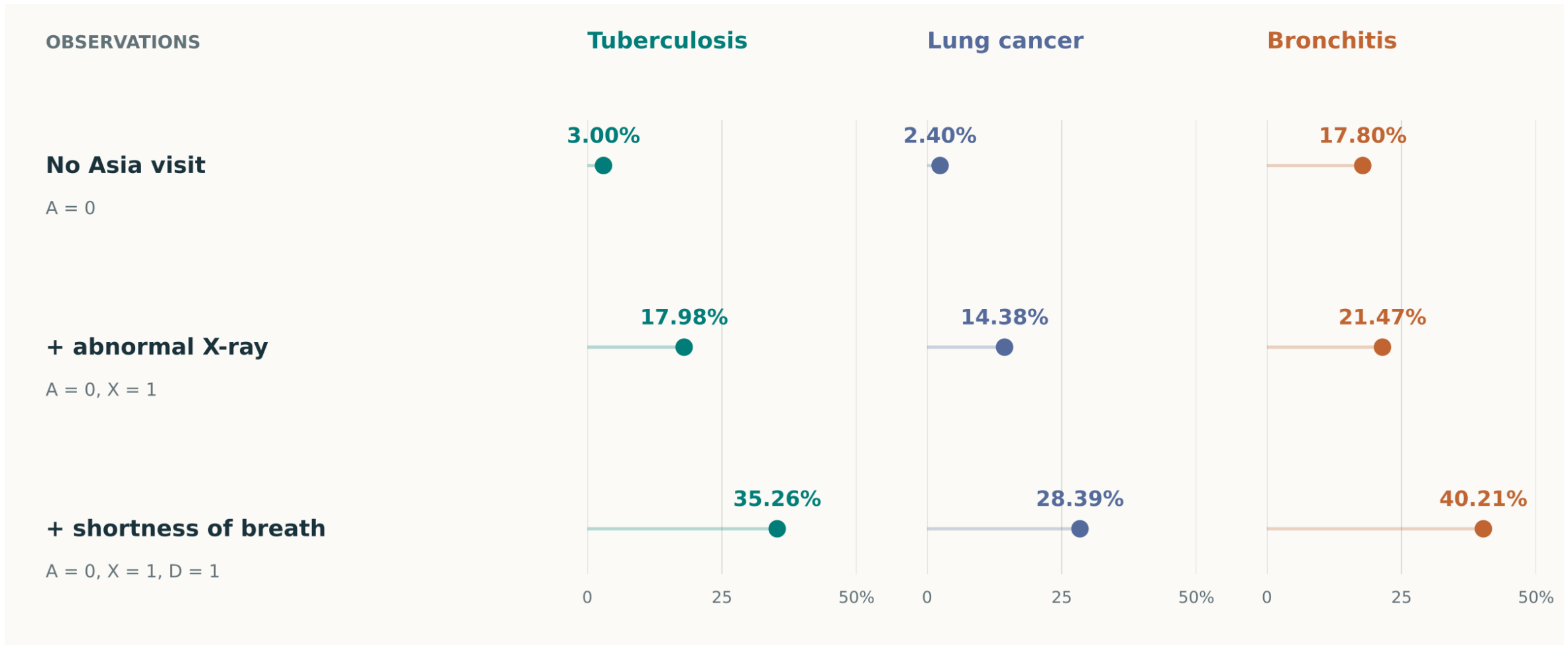
$$f_T(A, T) = P(T \mid A)$$



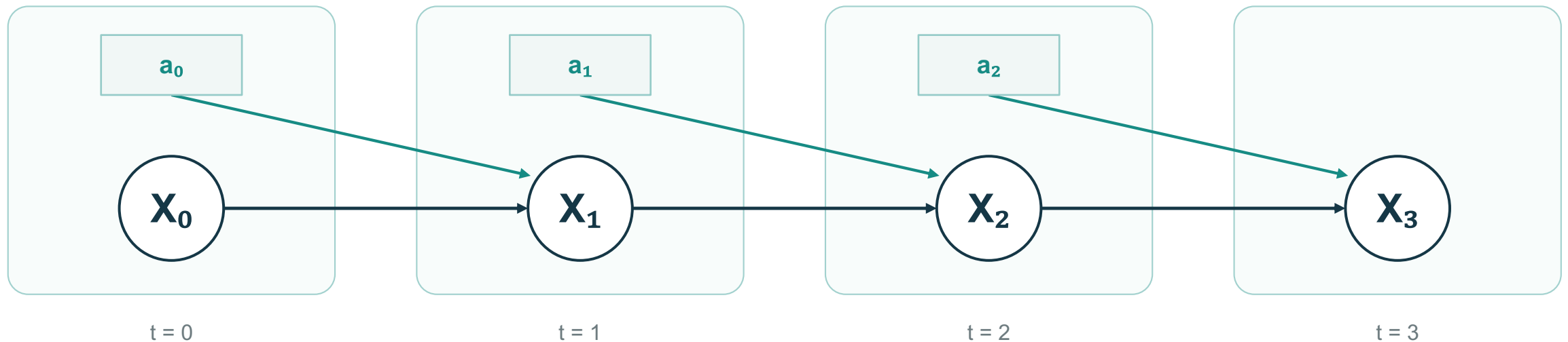
Zero-shot CPTs with Large Language Model (Jev)



Once CPTs are known, we can perform **inference**



***Dynamic Bayes nets* replicate a Bayes net fragment over time.**



Exercise: How many possible trajectories?

The room can repeat. The time index does not run backward.

Local tables avoid storing every possible state history.

ENUMERATE AN 11-STATE JOINT

48,828,125
 5^{11} entries



REUSE ONE LOCAL MOTION TABLE

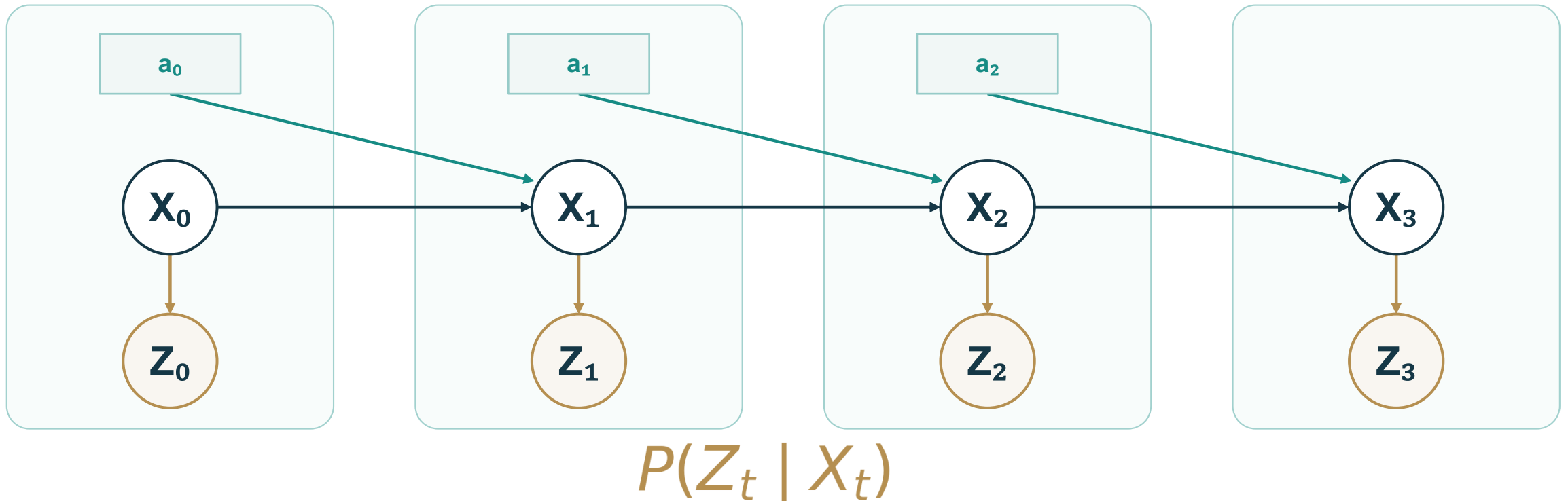
100 + 5

motion CPT + initial prior



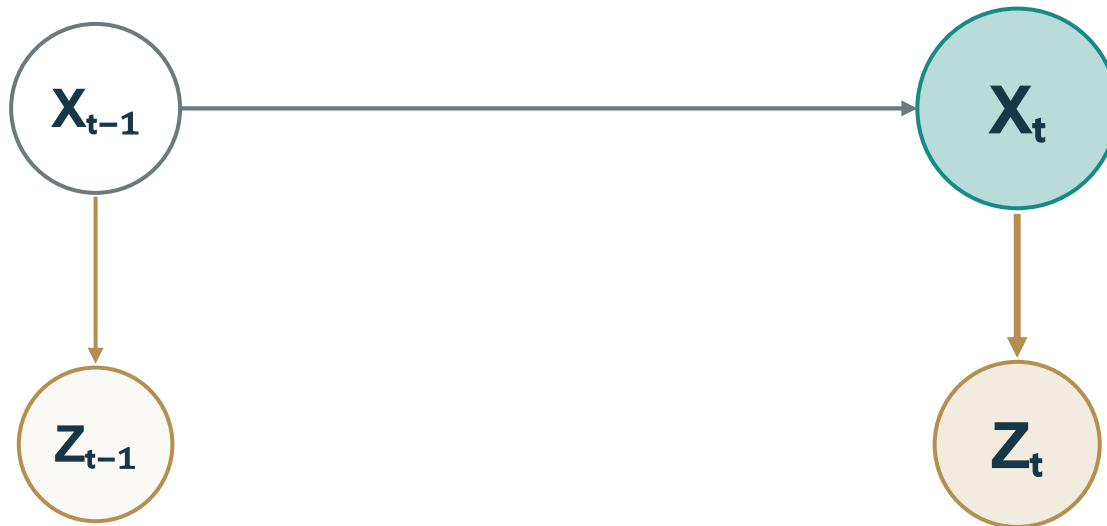
Same model at every step.

Each new state generates one new sensor reading.



Motion links one state to the next; sensing links each state to its own measurement.

The sensor ignores the past once the current room is known.

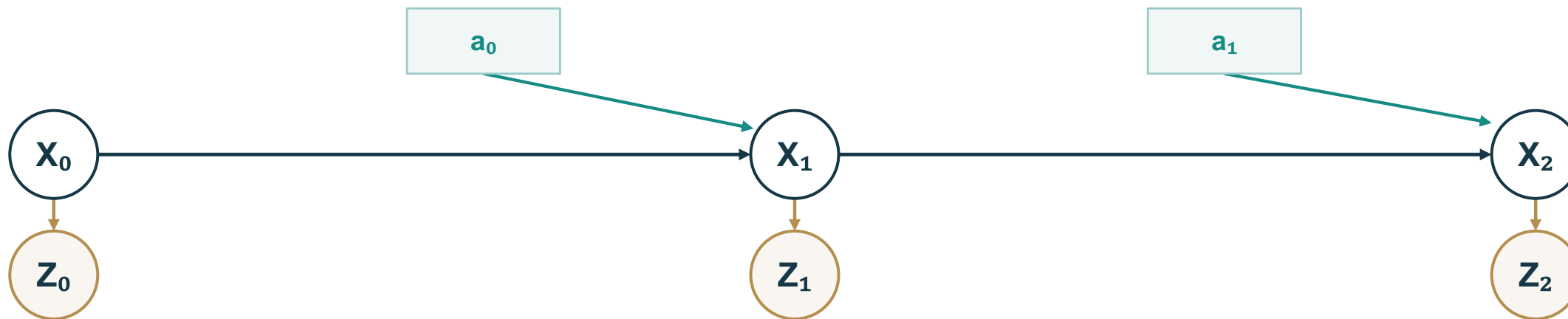


Condition on
this room.

Use only its
sensor distribution.

$$P(Z_t \mid X_{0:t}, Z_{0:t-1}, a_{0:t-1}) = P(Z_t \mid X_t)$$

The joint is a prior *times*
motion factors *times* sensor factors.



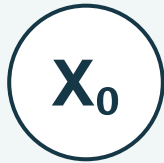
$$P(X_{0:2}, Z_{0:2} \mid a_{0:1}) = \overset{\text{prior}}{P(X_0)} P(X_1 \mid X_0, a_0) P(X_2 \mid X_1, a_1) \\ \times P(Z_0 \mid X_0) P(Z_1 \mid X_1) P(Z_2 \mid X_2)$$

Remember: Know quantities (actions in this case!) are indicated with a square node

How do we *simulate*?

First sample its parents, only then sample a node.

1 · INITIAL STATE



$$x_0 \sim P(x_0)$$

Start from the prior.

2 · MEASUREMENT



$$z_t \sim P(z_t \mid x_t)$$

Sample this room's light.

3 · NEXT STATE



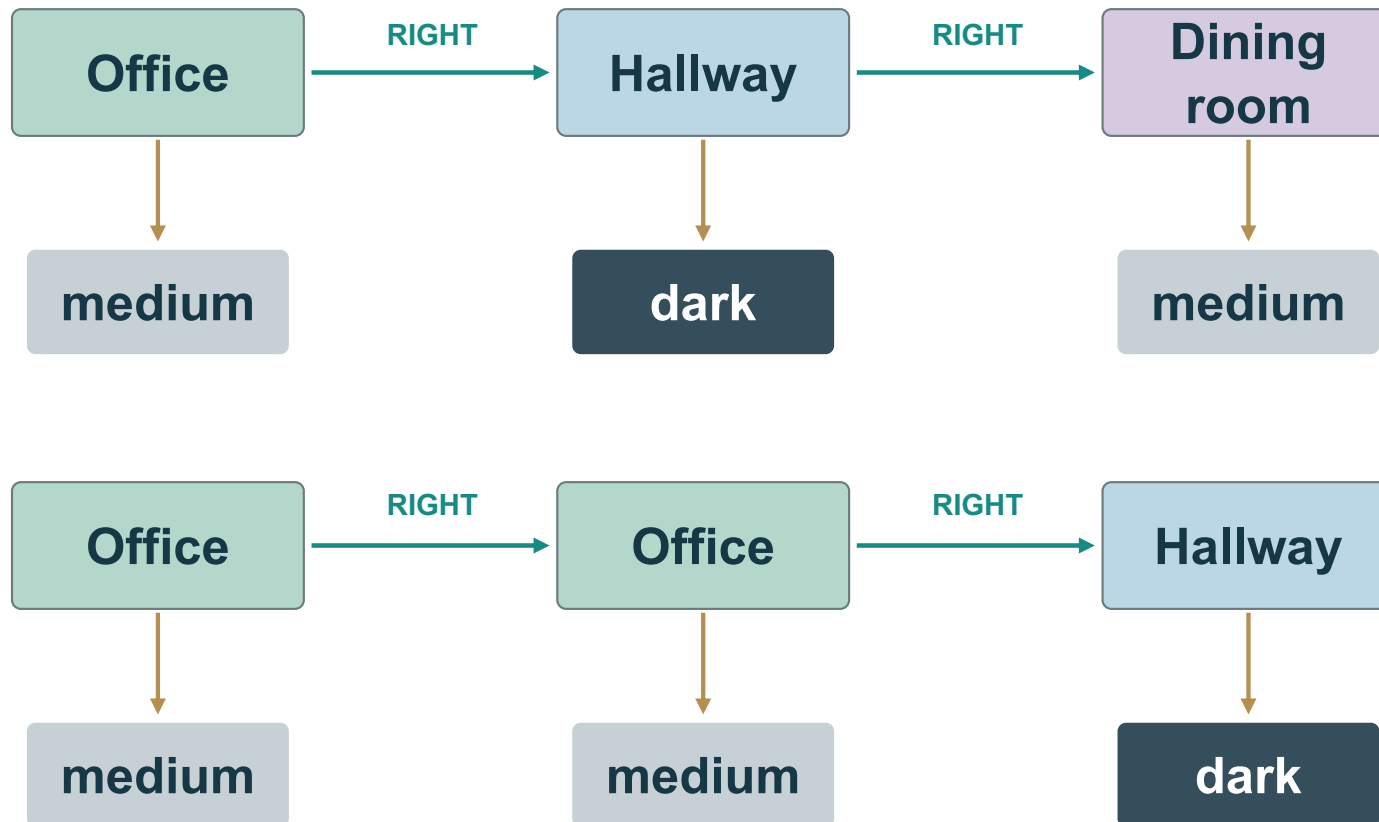
$$x_{t+1} \sim P(x_{t+1} \mid x_t, a_t)$$

Use the given action.

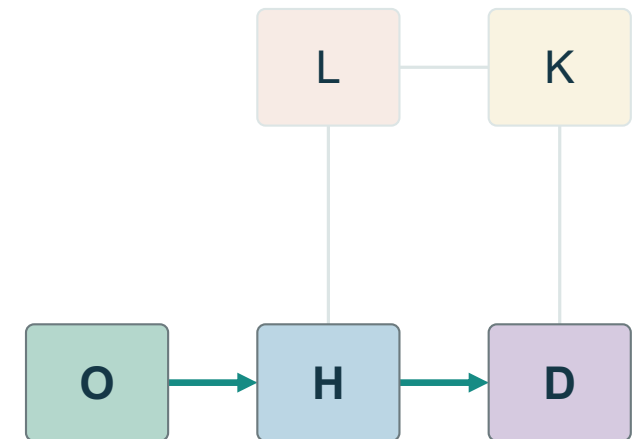
After advancing time, repeat measurement → next state. A *realization* is one possible world.

The same actions can produce different journeys and readings.

SAME START: OFFICE · SAME COMMANDS: RIGHT, RIGHT

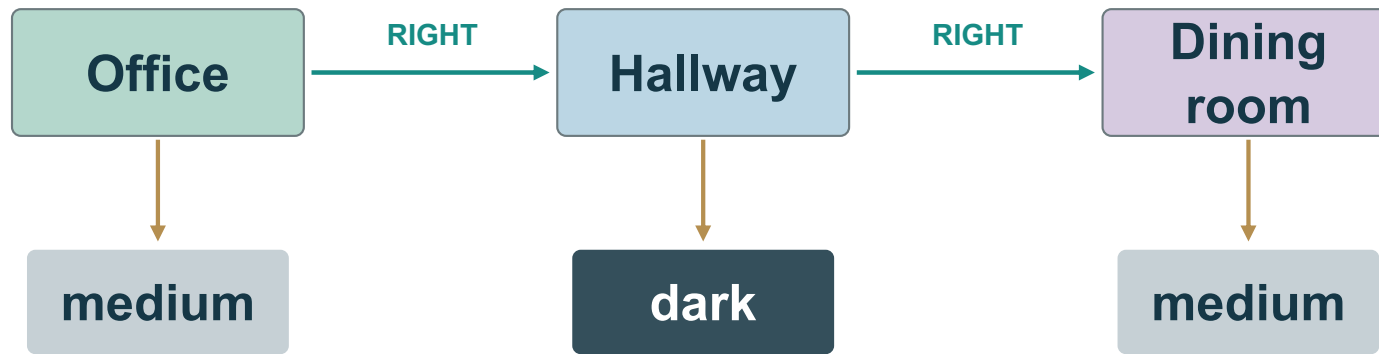


TWO POSSIBLE REALIZATIONS

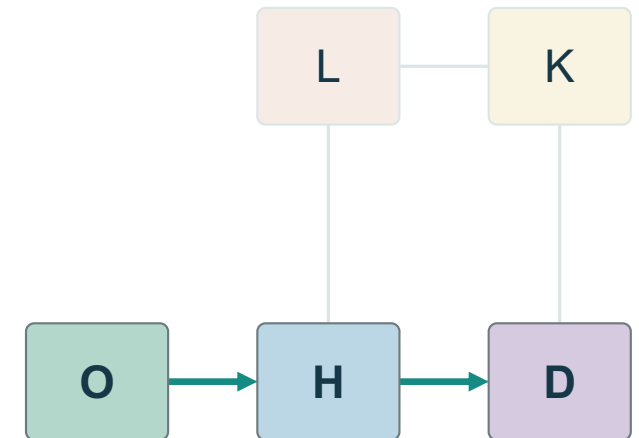


The same actions can produce different journeys and readings.

SAME START: OFFICE · SAME COMMANDS: RIGHT, RIGHT



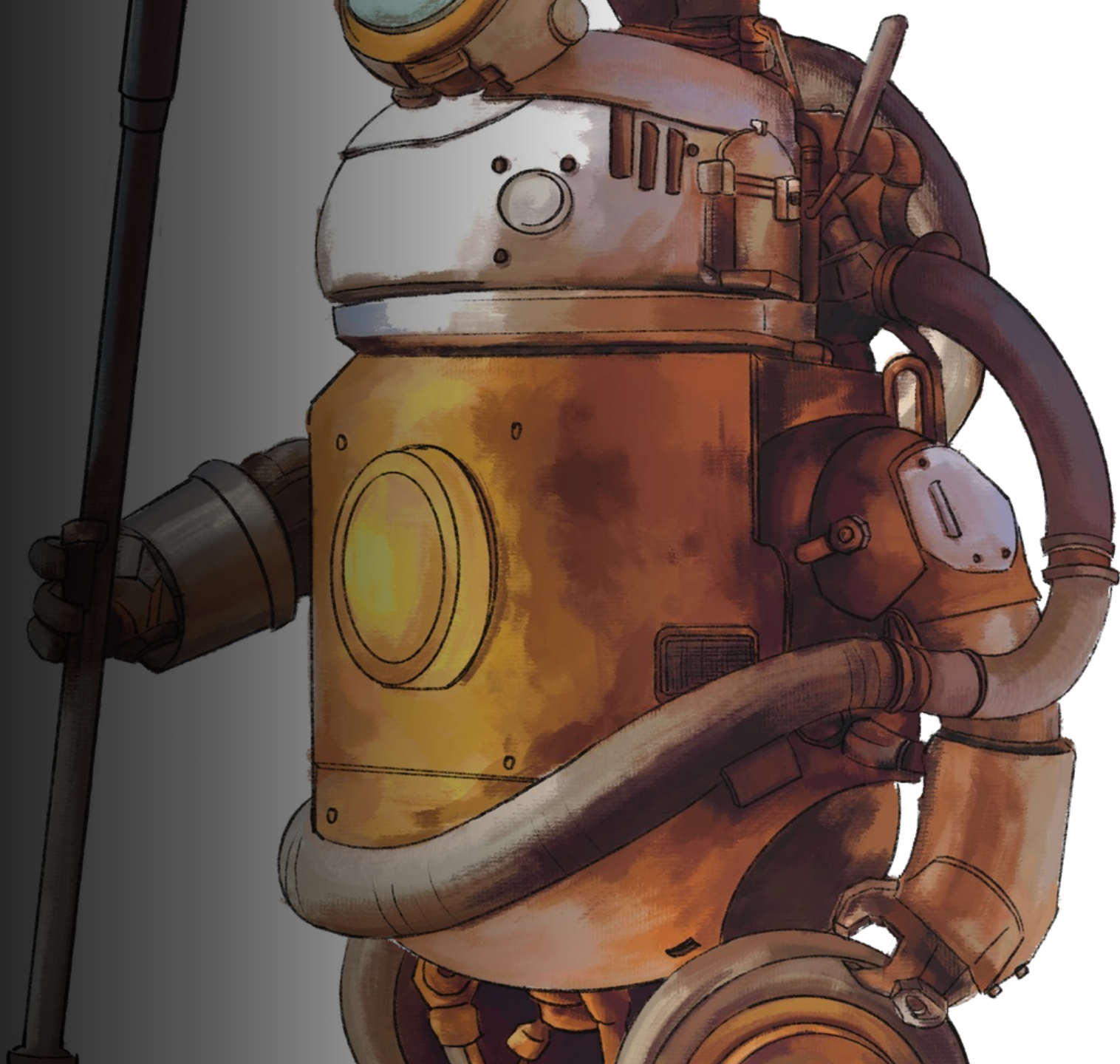
TWO POSSIBLE REALIZATIONS



Exercise: How many possible realizations?

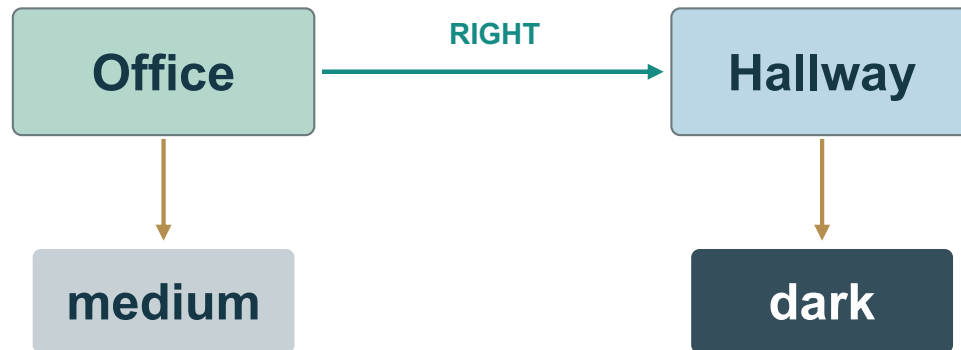


Perception



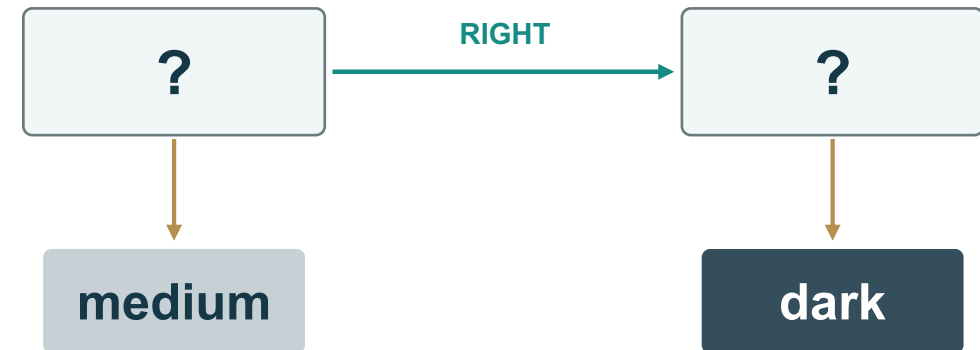
Simulation reveals the states. *Perception* must infer them.

SIMULATION: GENERATE A POSSIBLE WORLD



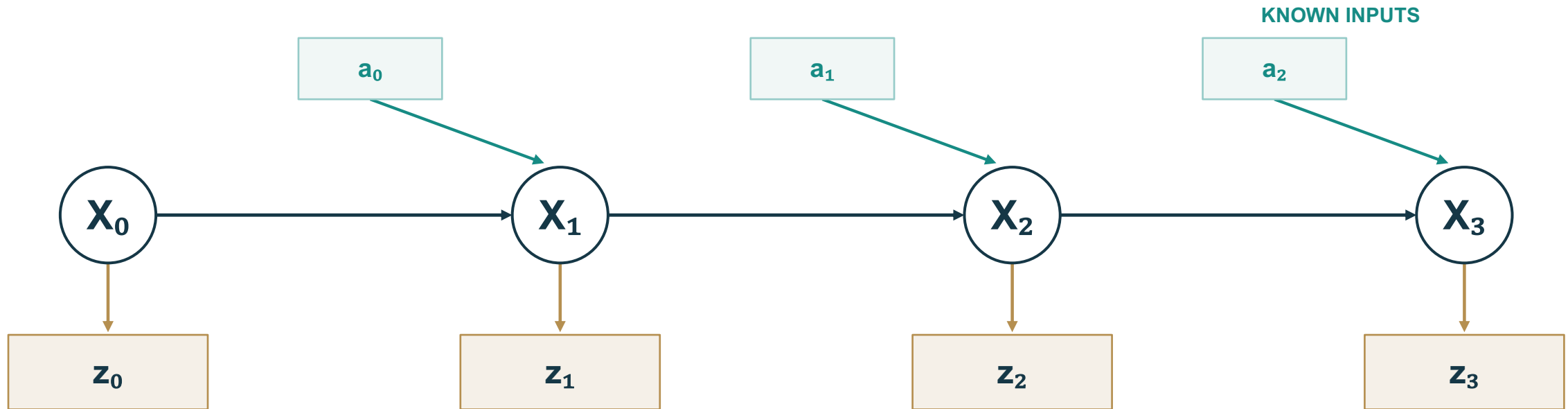
$$P(X, Z | a)$$

PERCEPTION: EXPLAIN THE KNOWN READINGS



$$P(X | z, a)$$

HMMs provide a general framework for perception over time.



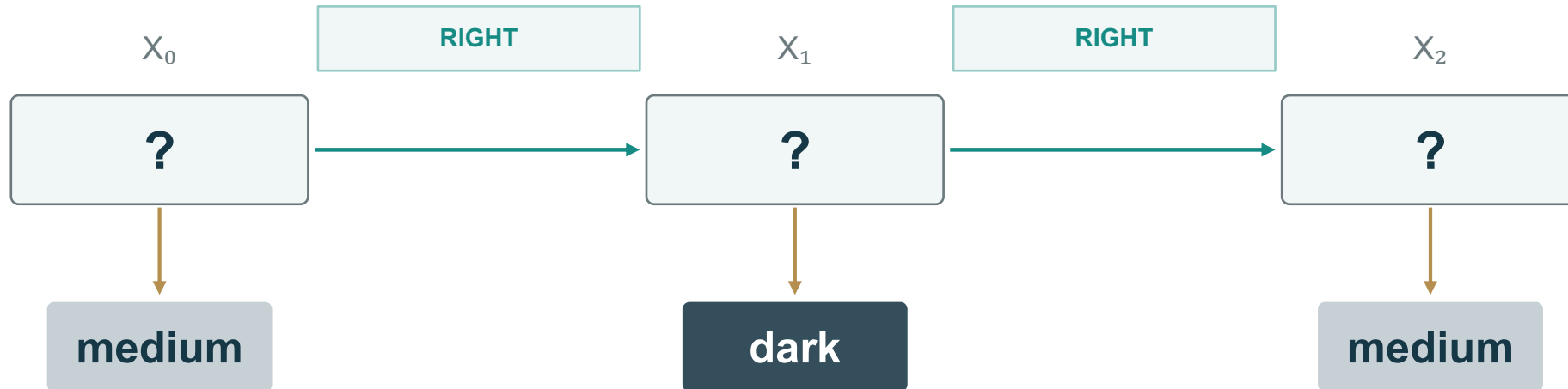
Hidden states: circles

Observed values: boxes

The state sequence is hidden; the actions and measurement values are given.

**We infer a state sequence,
not just the current room.**

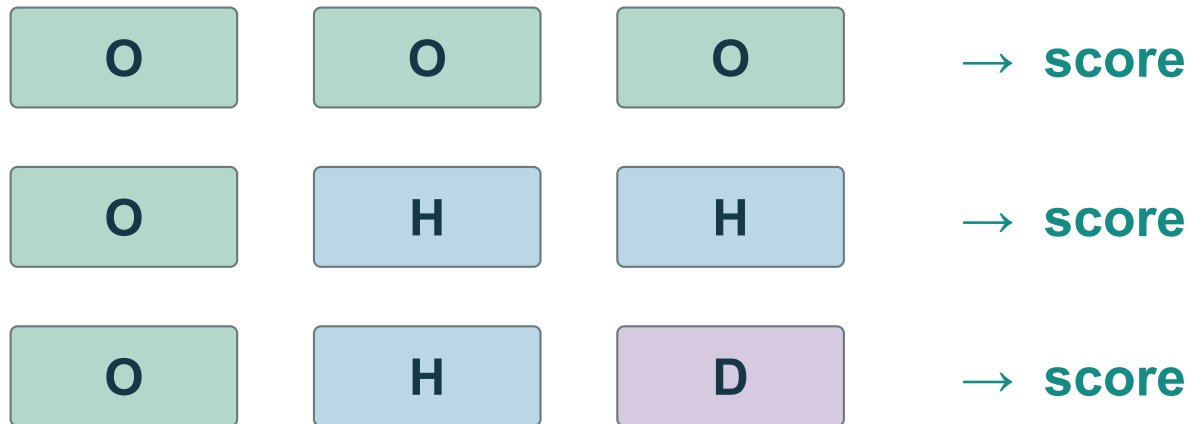
Unknown: the whole room history



$$P(X_0, X_1, X_2 \mid z_0, z_1, z_2, a_0, a_1)$$

Exercise: How many possible hypotheses?

**Fix the evidence, then compare
the joint scores of candidate paths.**



**Same evidence
Same actions
Same normalizer**

$$q(x) = P(X = x, Z = z \mid a)$$

$$P(X = x \mid z, a) = \frac{q(x)}{P(Z = z \mid a)}$$

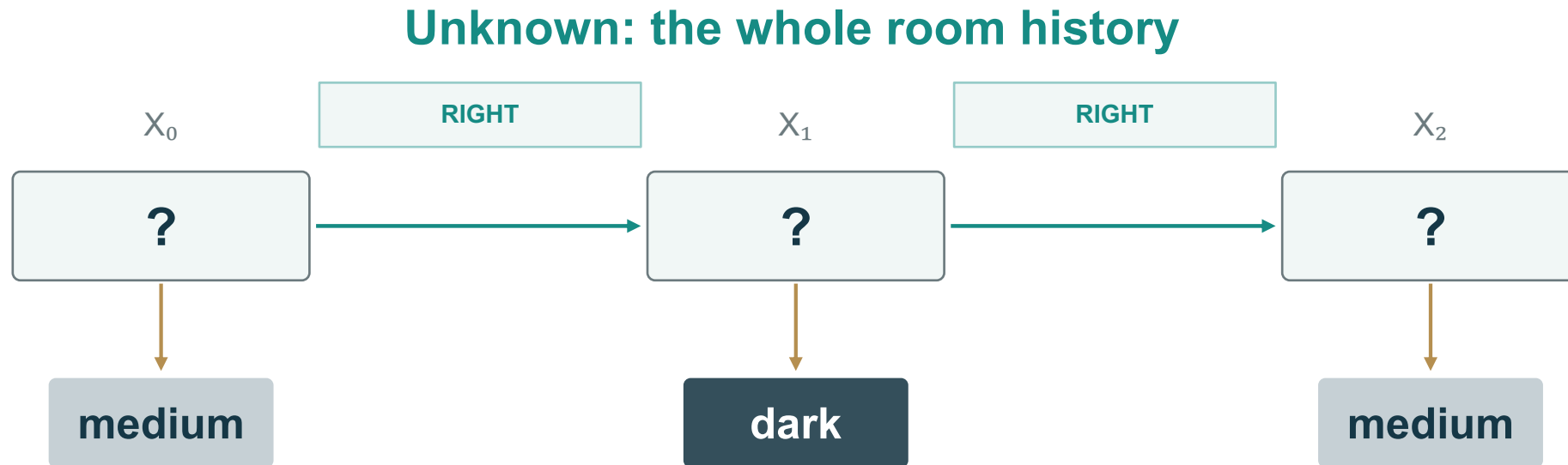
The most probable explanation is the highest-scoring complete path.



Illustrative ranking—not numerical scores yet.

$$x^* = \arg \max_x P(X = x \mid z, a) = \arg \max_x q(x)$$

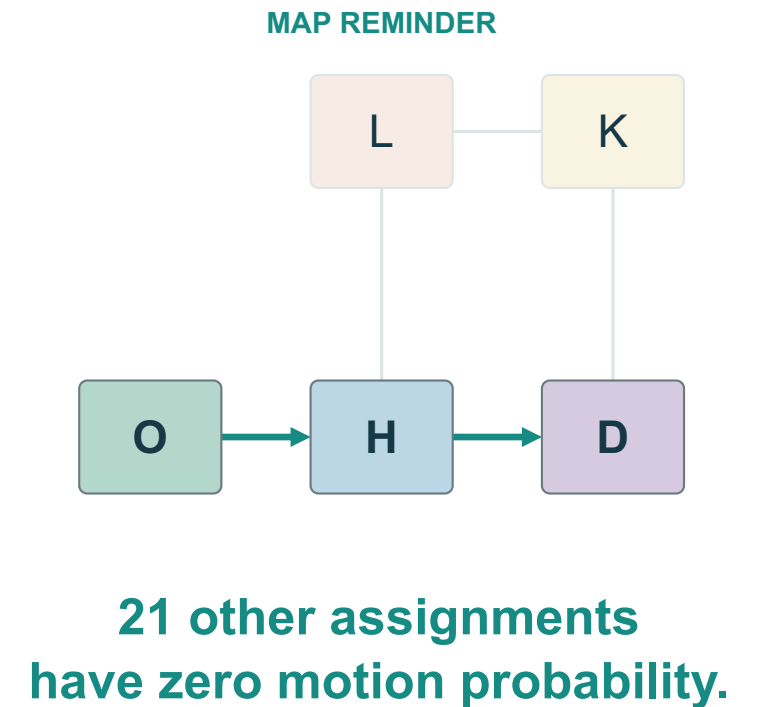
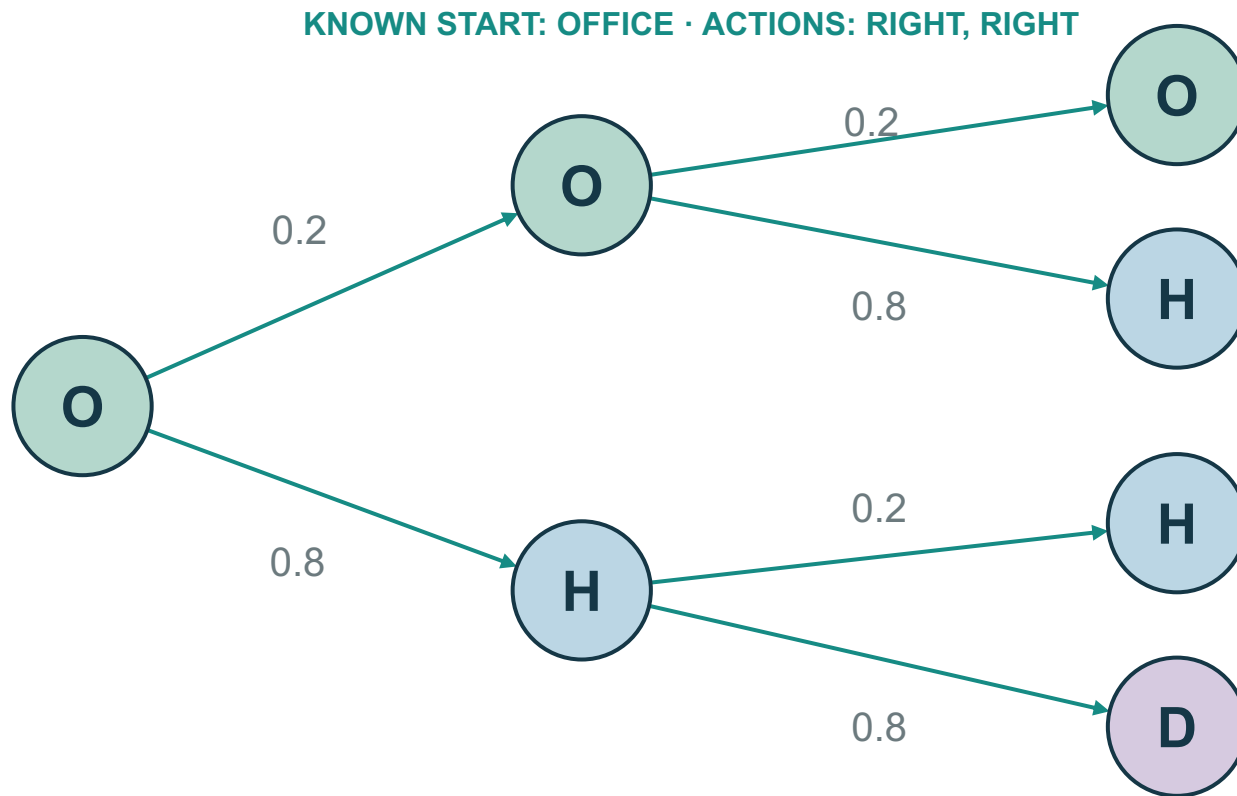
**We infer a state sequence,
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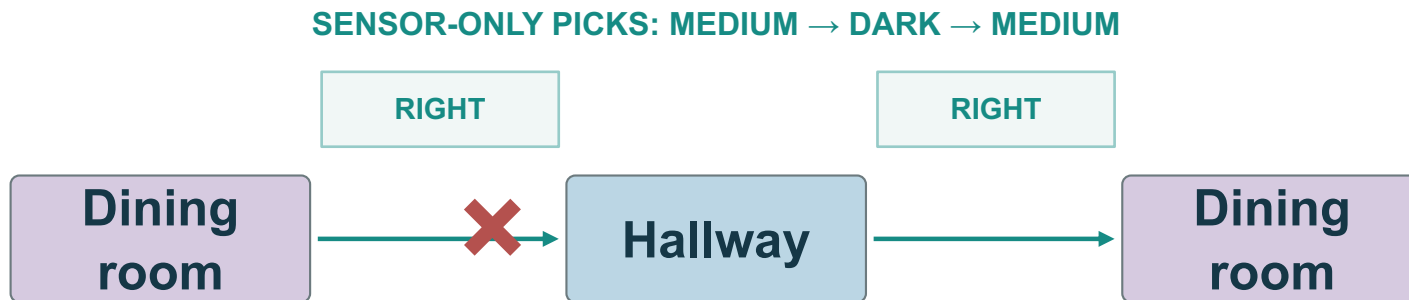
$$P(X_0, X_1, X_2 \mid z_0, z_1, z_2, a_0, a_1)$$

Exercise: How many of these have non-zero probability?

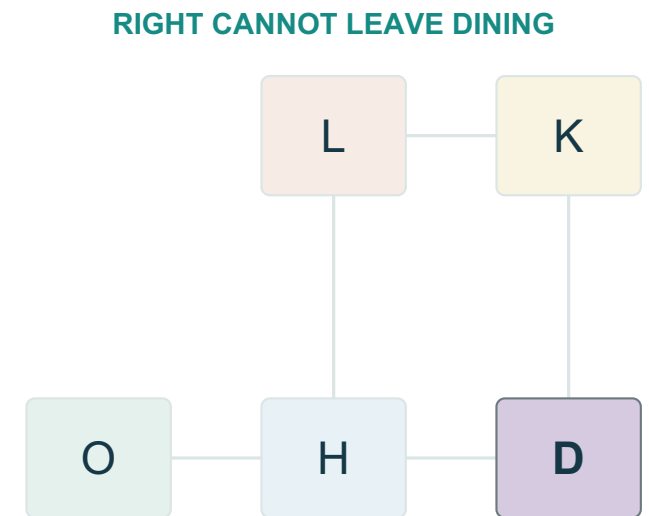
From the known Office start, only **four paths survive** two right commands.



The best room for each reading need not form a possible journey.



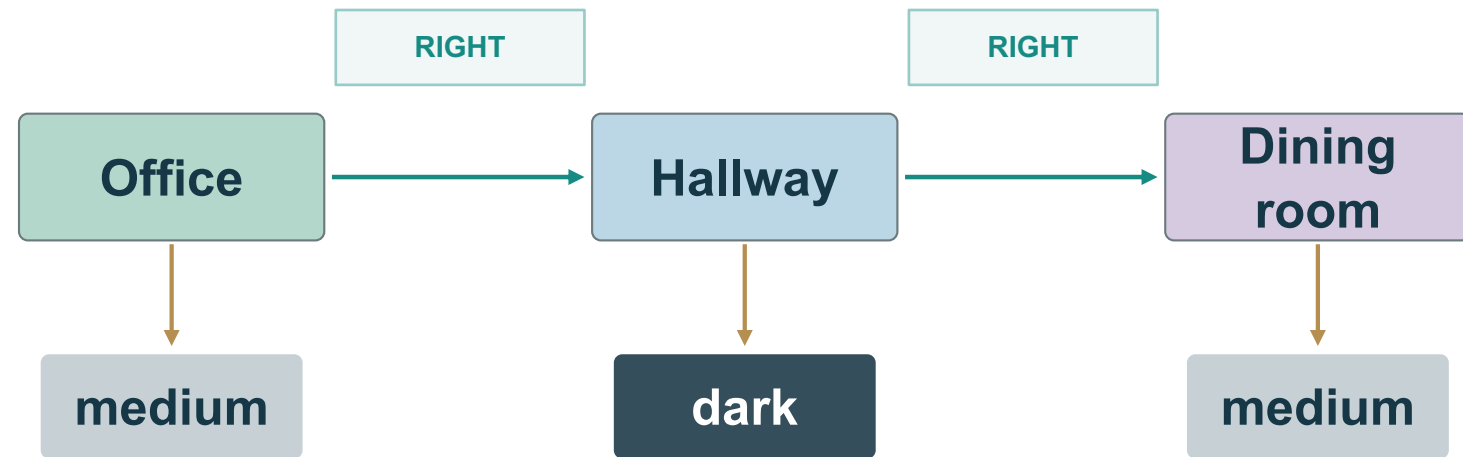
$$P(H \mid D, \text{right}) = 0$$



Compare complete paths—not three independent guesses.

A path's score multiplies its prior, motion, and sensor probabilities.

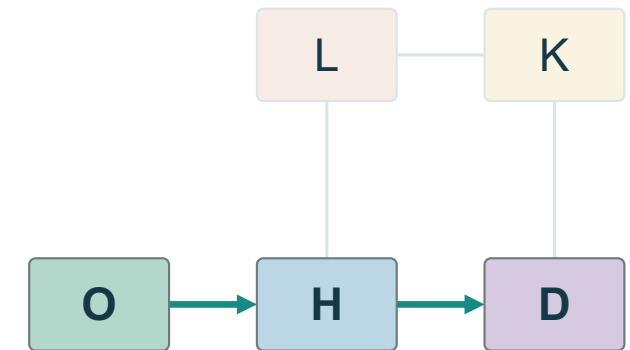
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Office	0.2	0.7	0.1
Hallway	0.8	0.1	0.1
Dining room	0.1	0.8	0.1



$$\begin{array}{ccccccc}
 \boxed{1} & \times & \boxed{0.7} & \times & \boxed{0.8} & \times & \boxed{0.8} & \times & \boxed{0.8} & \times & \boxed{0.8} & = & \boxed{0.28672} \\
 \text{prior} & & \text{sensor} & & \text{motion} & & \text{sensor} & & \text{motion} & & \text{sensor} & &
 \end{array}$$

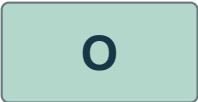
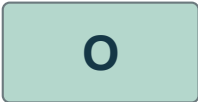
Joint score—not a posterior percentage.

KNOWN START: OFFICE

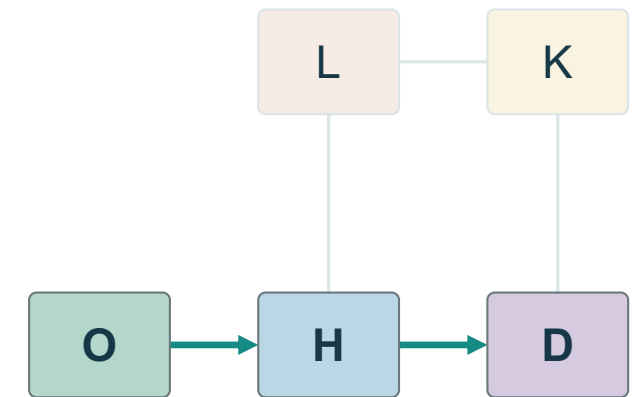


Office → Hallway → Dining room has the highest joint score.

KNOWN START: OFFICE · READINGS: MEDIUM, DARK, MEDIUM

Candidate path	Joint score $q(x)$
 →  → 	0.00392
 →  → 	0.00224
 →  → 	0.00896
 →  → 	0.28672

MOST PROBABLE EXPLANATION



Same prior, actions, and measurements for every candidate.

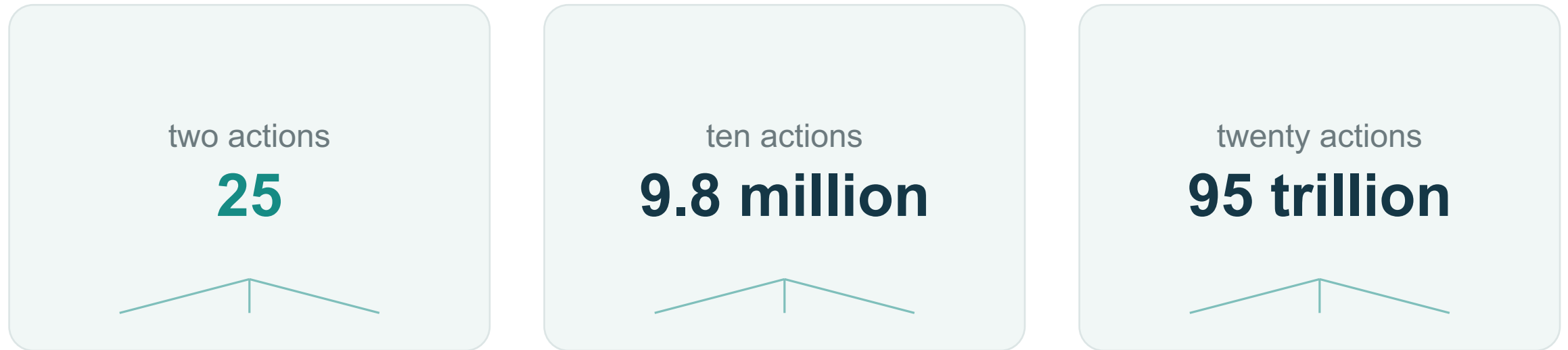
The best explanation is likely, not certain.



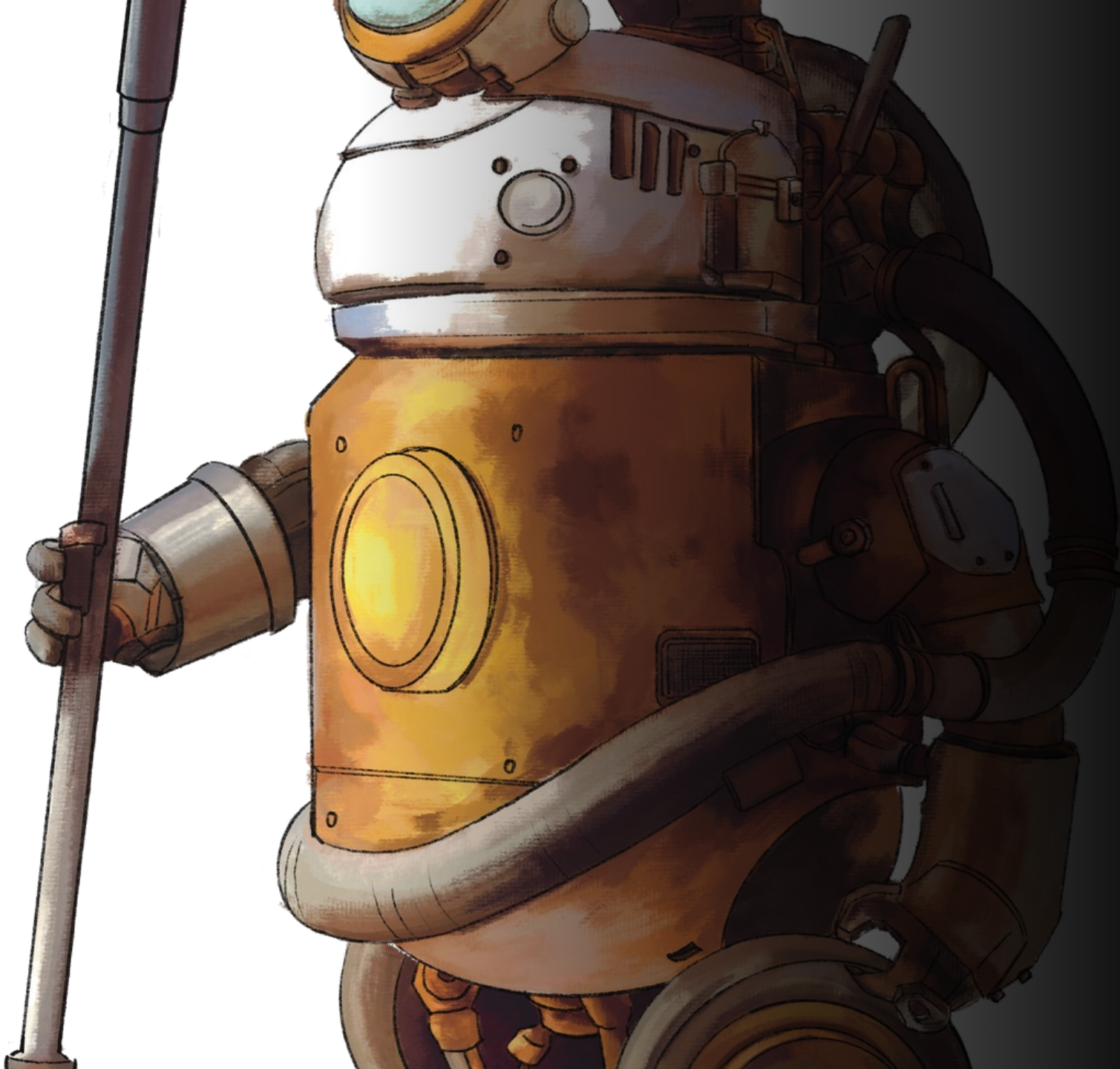
$$P(x \mid z, a) = \frac{q(x)}{0.30184}$$

The remaining paths still carry 5.01%.

Scoring one path is easy;
enumerating all paths is intractable in general !

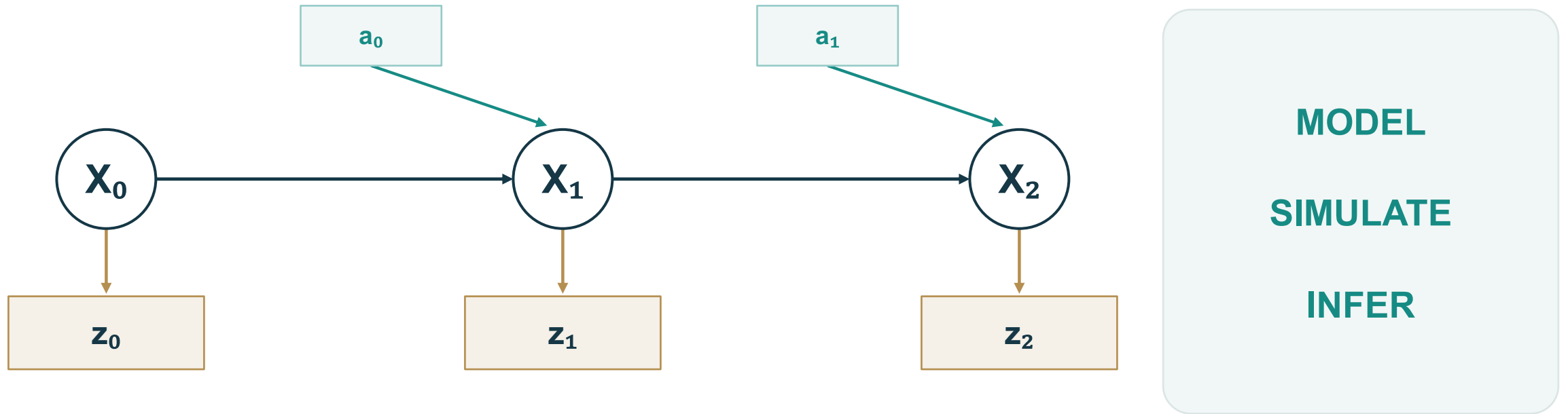


5^T candidate assignments (X_0 fixed)



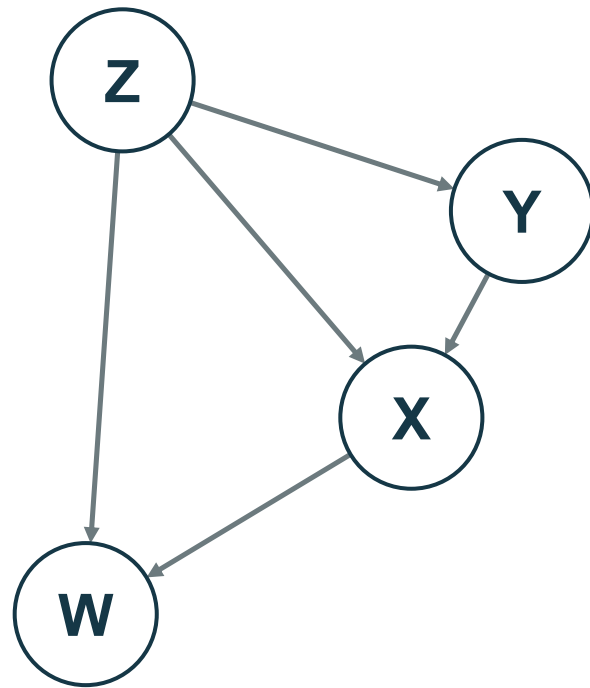
Takeaways

**A motion model links states;
a sensor model links states to evidence.**

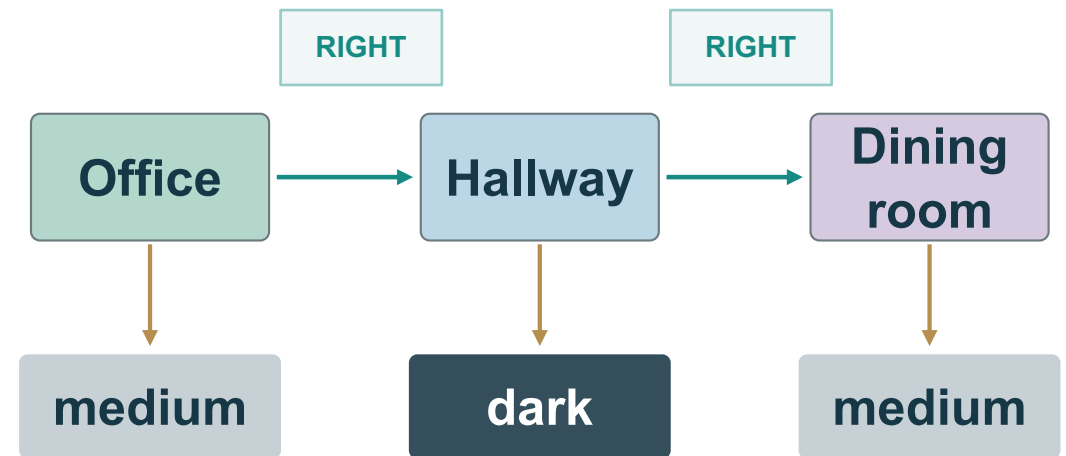


Prior $\times$ motion $\times$ sensing $\rightarrow$ joint model $\rightarrow$ an explanation of the observed journey.

Bayes nets model the journey. Inference explains the measurements.



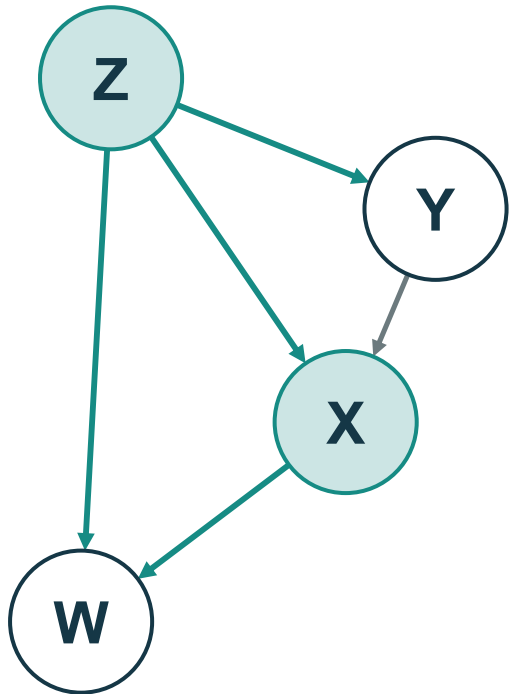
A factored generative model



A likely hidden-state sequence

Next question: how can we avoid enumerating every path?

The chain rule explains why the Bayes-net product works.



CHAIN RULE

$$P(W, X, Y, Z) = P(W \mid X, Y, Z)P(X \mid Y, Z)P(Y \mid Z)P(Z)$$

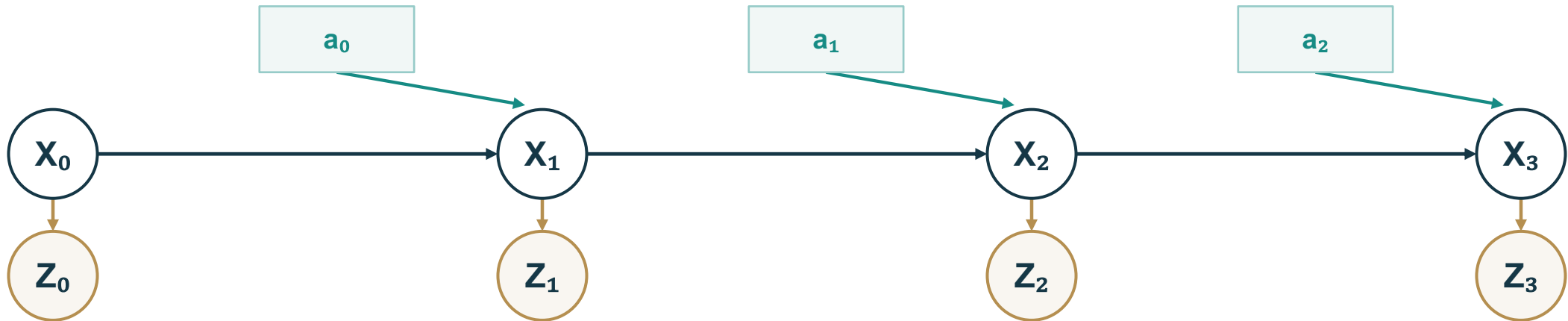
USE THE GRAPH

$$P(W \mid X, Y, Z) = P(W \mid X, Z)$$

SUBSTITUTE

$$P(W, X, Y, Z) = P(W \mid X, Z)P(X \mid Y, Z)P(Y \mid Z)P(Z)$$

The full sequence uses T actions and $T+1$ states and measurements.



$$P(X_{0:T}, Z_{0:T} \mid a_{0:T-1}) =$$

$$P(X_0) \prod_{t=0}^{T-1} P(X_{t+1} \mid X_t, a_t) \prod_{t=0}^T P(Z_t \mid X_t)$$